

Roll No.

TEE-608

B. TECH. (EE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
HIGH VOLTAGE ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define Townsend's first and second ionization coefficients. How is the condition for breakdown obtained in a Townsend discharge ?

(CO3)

OR

- (b) Describe the current growth phenomenon in a gas subjected to uniform electric field.

(CO3)

2. (a) Explain the difference between photoionization and photoelectric emission.

(CO2)

OR

- (b) How is the electric field intensity controlled ?

(CO2)

3. (a) What is meant by control of transient or impulse voltages ? Why is this necessary ?

(CO1)

P. T. O.

(2)

OR

(b) What are surge voltages and how are they distributed in windings of a transformer ? (CO1)

4. (a) What are commercial liquid dielectrics and how are they different from pure liquid dielectrics ? (CO1)

OR

(b) Explain the phenomenon of electrical conduction in liquids. How does it differ from that in gases ? (CO1)

5. (a) How does the internal discharge phenomenon leads to breakdown in solid dielectrics ? (CO4)

OR

(b) Explain the different mechanisms by which breakdown occurs in solid dielectrics in practice. (CO4)

Roll No.

TEE-605

B. TECH. (EE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
DIGITAL CONTROL SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by digital control ? Explain the process of conversion of an analog signal into a digital signal. (CO1)

OR

- (b) List the various advantages of digital control over analog control. Also, discuss the effects of the sampling and quantization on implementation of digital control. (CO1)

2. (a) What are various types of representations of a discrete time system ? Explain any *one* in detail. (CO1)

OR

- (b) What are the various basic discrete time sequences used in digital control ? Explain each *one* in brief. (CO1)

P. T. O.

(2)

3. (a) Solve the following difference equations using z-transform : (CO2)

(i) $y(k+2) + 3y(k+1) + 2y(k) = 0; y(-1) = -0.5, y(-2) = 0.75$

(ii) $y(k) - 3y(k-1) + 2y(k-2) = r(k); k \geq 0$

Consider $r(k)$ to be a unit step sequence. (CO2)

OR

- (b) List the various methods to map a system from s-plane to z-plane. Explain any *one* method in detail with proper derivation. (CO2)

4. (a) Describe the Jury Stability Criterion. Also, discuss the necessary and sufficient conditions for stability used in Jury stability criterion. (CO2)

OR

- (b) Determine the stability of the systems described by the following characteristic equations (use Jury Stability Criterion) : (CO2)

(i) $Z^3 - 1.3z^2 - 0.08z + 0.24 = 0$

(ii) $z^4 - 1.368z^3 + 0.4z^2 + 0.08z + 0.002 = 0$

5. (a) An analog system is expressed by the transfer function $G(s) = \frac{10}{s(s+2)}$.

Obtain the discrete equivalent of the above system using bi-linear transformation. (CO2)

OR

- (b) Derive the relationship between s-plane and z-plane using step invariance method. (CO2)

Roll No.

TOE-604

B. TECH. (EE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
COMMUNICATION ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Derive mathematically and draw amplitude spectrum of an amplitude modulated waveform. (CO1)

OR

- (b) Draw the block diagram of digital communication system and explain all its components. (CO1)

2. (a) Explain communication system and its components with the help of a block diagram. (CO1)

OR

- (b) Derive mathematical function for a trapezoidal signal whose base starts at origin and ends at 10 units. Its height is 3 units. Its upper flat side is of 6 units and starts from 2 units. (CO1)

P. T. O.

(2)

3. (a) Discuss the generation of DSB-SC waveform using balanced modulator method. (CO1/CO2)

OR

- (b) What is over modulation in case of Amplitude Modulation ? Discuss with help suitable waveforms. (CO1/CO2)
4. (a) Prove that $\frac{2}{3}$ rd power can be saved in case of DSB-SC modulation in comparison to DSB-FC modulation, assuming 100% modulation. (CO1/CO2)

OR

- (b) Design a square law de-modulator for AM detection and discuss its working. Derive the required mathematical expression for getting message signal at the receiver end. (CO1/CO2)
5. (a) Derive mathematical expression for AM waveform assuming a sinusoidal carrier waveform. Also draw message signal, carrier signal and AM waveform. (CO2)

OR

- (b) Write short notes on the following : (CO2)
- (i) SSB-SC
 - (ii) Frequency modulation

Roll No.

TEE-603

B. TECH. (EE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
POWER SYSTEM PROTECTION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) "Relay is the brain of protection." Justify this statement using trip circuit. (CO1)

OR

- (b) Describe the essential features of a protective relay with reference to reliability, selectivity speed of operation and discrimination. (CO1)
2. (a) Discuss the concept of dividing the system into protective zones. Draw suitable diagram to support the explanation. (CO1)

OR

- (b) Analyze the transient behavior in Synchronous machine and discuss how overall reactance increase with time. (CO1)
3. (a) Define the term Pick up value as used in the context of an overcurrent relay. What is the significance of TMS ? (CO2)

P. T. O.

(2)

OR

- (b) The current setting is set at 75%, TMS = 0.4; C.T. Ratio = 400/5, fault current is 6000 A. Determine the operating time of the relay. At TMS = 1 the operating time at various PSMs are shown in the table below :

PSM	Operating time (in secs.)
2	10
4	5
5	4
8	3
10	2.8
20	2.4

(CO2)

4. (a) With a neat diagram, discuss its working principle of Buchholz relay.

(CO2)

OR

- (b) A transmission of inductance 0.1 H and resistance 5 ohm is suddenly short-circuited at the far end. Write the expression for short circuit current $i(t)$. Consider $v = 100 \sin(100\pi t + 15)$. (CO2)
5. (a) Explain the distance relay protection scheme. How are they superior to overcurrent relays ? (CO3)

OR

- (b) What is a differential relay ? What are its applications ? (CO3)

Roll No.

TEE-602

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

POWER SYSTEM-II

Time : 1½ Hours

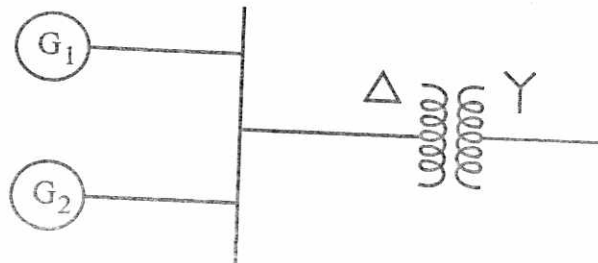
Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you mean by reactance diagram of the power systems ? Draw reactance diagram of the power systems given in figure. Assume base MVA is 60 MVA.

(CO1)



P. T. O.

The rating of generators and transformer are indicated in Table 1 : (CO1)

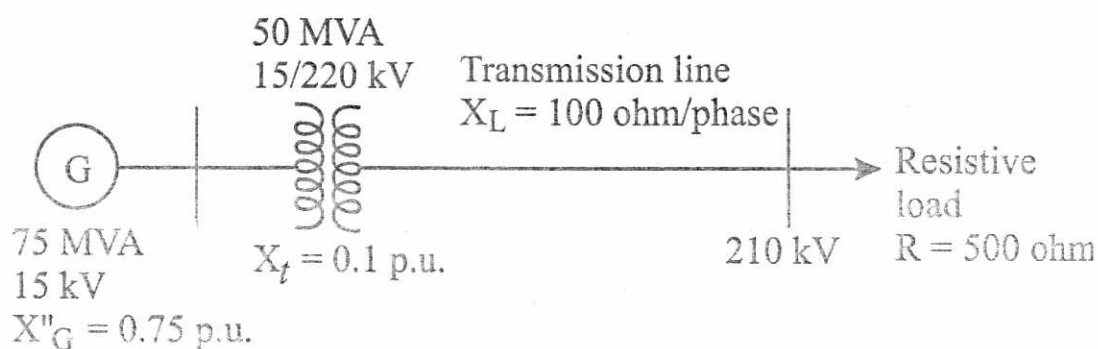
	Ratings	Reactance (in p. u.)
Generator 1 (G_1)	30 MVA, 11 kV	0.2
Generator 2 (G_2)	25 MVA, 11 kV	0.25
Transformer	60 MVA, 11 kV/66 kV	0.10

OR

- (b) A 15 kVA 200/400 V, 50 Hz single phase transformer has primary and secondary leakage reactance each of 5Ω . Determine total reactance in p.u. (CO1)
2. (a) Discuss the conditions for the existence of zero sequence component of the current in Y- and Δ -connected winding. Also, show that neutral current exists in the zero-sequence network, only. (CO2)

OR

- (b) Derive three phase complex power in terms of symmetrical components for three phase unbalanced power systems. (CO2)
3. (a) For the system, as shown in the figure, determine p. u. value of all the reactance on common base. (Assume common base values are 100 MVA and 15 kV for the primary side of the transformer). (CO2)



OR

- (b) A balance star connected load takes 150 A from a balanced 3-phase 4-wire supply. If the fuses in two of the lines are removed, find the symmetrical components of the line current before and after the fuses are removed. (CO2)
4. (a) Apply symmetrical component analysis and interconnect the sequence networks in case of LLG faults. (CO3)

OR

- (b) In a three-phase four wire system the currents in the lines a , b and c under abnormal conditions of loading were as follows : (CO3)

$$I_a = 50 \angle 45^\circ$$

$$I_b = 40 \angle 300^\circ$$

and

$$I_c = 30 \angle 180^\circ$$

Calculate the zero, positive, and negative phase sequence currents in line a and the return current in the neutral conductor.

5. (a) Show that the voltage of healthy phase w.r.t. ground in case of LG fault in phase 'a' is given as : (CO3 & CO1)

$$V_b = \frac{Z_0(\alpha^2 - 1) + Z_2(\alpha^2 - \alpha)}{Z_0 + Z_1 + Z_2}$$

P. T. O.

OR

- (b) A three phase 10 MVA, 11 kV synchronous generator with the solidly grounded neutral point supplies a feeder. The relevant impedance of the generator and feeder are as follows : (CO3 & CO1)

	Generator	Feeder
Z_1	$j 1.2$	$j 1.0$
Z_2	$j 0.9$	$j 0.9$
Z_3	$j 0.4$	$j 3.0$

A fault from one phase to ground occurs at the far end of the feeder. Determine the voltage to neutral of the faulty phase at the terminals of the generator.

Roll No.

TEE-601

B. TECH. (EE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

POWER ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain all the modes of operation of SCR. Also Draw its V-I characteristics. Define the Latching Current and Holding Current of SCR. (CO1, CO2)

OR

- (b) With the help of waveform, explain the turn on characteristics of SCR. (CO1, CO2)

2. (a) Explain the protection methods of SCR. (CO1, CO2)

OR

- (b) With the help of diagram, explain the operation of MOSFET. (CO1, CO2)

P. T. O.

3. (a) An SCR with heat Sink arrangement has ambient temperature 50 degree Celsius and junction temperature 100 degree Celsius. Average power dissipated is 50 W. If junction temperature is reduced to 70 degree Celsius, calculate the percentage reduction in thermal resistance.

(CO1, CO2)

OR

- (b) What do you understand by snubber circuit and its requirement during turn on and turn off the SCR operation during series operation ?

(CO1, CO2)

4. (a) Explain the single phase half wave control rectifier for RLE load with suitable waveform and circuit diagram. Calculate its average output voltage.

(CO1, CO2)

OR

- (b) In the single phase 230 V, 50 Hz AC main fed step down transformer with turn ratio (4 : 1) is supplying power to half wave uncontrolled a.c.-d.c. converter used for charging a battery of 12 V with the series limiting resistor of 19.04 ohm. Calculate the average charging current required to charge the battery.

(CO1, CO2)

(3)

5. (a) In the single phase 230 V, 50 Hz, fully controlled converter feeding to electric braking of a separately excited DC motor. The DC motor load is represented with resistance 2 ohm and back e.m.f. 150 V. Assume that load inductance is sufficient to ensure continuous and ripple free load current. Calculate the firing angle of the bridge for load current of 10 A. (CO1, CO2)

OR

- (b) Explain the three-phase half wave controlled rectifier with firing angle 60 degree. Also calculate the average output voltage. (CO1, CO2)